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Medicine model

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Medicine concentrations

Simulate how the medicine in a pill someone takes enters into the bloodstream, does its work and is filtered out again.

Medicine model

In the following equations the quantities of medicine in the GI tract at time t are given by $M_{GI}(t)$ and $M_{blood}(t)$ respectively. V_{GI} is the volume of the contents in the GI tract and V_{blood} is the volume of blood of the person taking the medicine.

$$\begin{aligned}\frac{dM_{GI}}{dt} &= -d_1 \left(\frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) \\ \frac{dM_{blood}}{dt} &= d_1 \left(\frac{M_{GI}(t)}{V_{GI}} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}}\end{aligned}$$

Simplified medicine model

Take the concentration of the medicine in the stomach $C_{GI} = \frac{M_{GI}}{V_{GI}}$ constant. The model then reduces to:

$$\frac{dM_{blood}}{dt} = d_1 \left(C_{GI} - \frac{M_{blood}(t)}{V_{blood}} \right) - d_2 \frac{M_{blood}(t)}{V_{blood}}$$

Tasks for the project in Module 3

1. Keep trying to collect quantitative information on your populations. If you cannot find that information, make an educated guess. Do not forget to write down where you found the information!
2. Throughout, record your findings. Write down the values you used, store the graphs with descriptive names and write down all conclusions you reached. Store the different versions of the program with descriptive names as well, so you can easily rerun a program later.
3. Start with the simplified model. Choose a value for C_{GI} and an initial condition $M_{blood}(0)$ (for example 0).
4. Implement Euler's method for the simplified model. Start with $d_2 = 0$. Decide on the range of values for d_1 that you want to test. Use the highest value of that range for d_1 to do the stepsize experiments with. (Why the highest value?) Simulate $M_{blood}(t)$ with different stepsizes. Estimate the error, using the result with the doubled stepsize. Decide how precise you want to calculate $M_{blood}(t)$, and choose a stepsize accordingly.
5. Then set $d_1 = 0$ and $M_{blood}(0)$ to some high number and simulate how the medicine is cleared from the blood. Decide on the range of values for d_2 that you want to test. Use the highest value of that range for d_2 to do the stepsize experiments with. Calculate the error again, and see whether that error is small enough for you. Choose a stepsize accordingly.
6. Run some experiments for different (combinations of values of d_1 and d_2). Try out how sensitive the results are to changes in the parameters. Choose values that fit the quantitative information you have, or that give results that seem reasonable. Do not forget to use the units for d_1 and d_2 when you write it down.

When you have completed the theoretical part of module 3, you can start already on the full basic model. You can also choose to leave the next points for module 4.

7. Implement Euler's method for the system for $M_{GI}(t)$ and $M_{blood}(t)$. If you have decided on different stepsizes for the calculations in items 4 and 5, for the system you need to calculate with the minimum of the two stepsizes. If in doubt check the errors again.

8. Check again the values of d_1 and d_2 . Are the results plausible?

Communicate well with your teammate, so you make the different choices together. Document every result and decision in your logbook. That will help in the communication between you two, and when you are writing the report later on, you can find all your previous results easily.

The next page is for the questions you might have about the project.

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